

Shahrad Zomorodi

✉ shahradzomorodi@gmail.com ☎ (949)-527-8214 🔗 [linkedin.com/in/shahradzomorodi](https://www.linkedin.com/in/shahradzomorodi)

EDUCATION

University of California, San Diego

B.S. Mechanical Engineering

Expected March 2027

Sept. 2024 – Present

COURSEWORK

Courses: Space Mission Analysis & Design, Heat Transfer, Fluid Mechanics (Advanced), Thermodynamics, Mechanics of Materials, Dynamics & Vibrations, Computational Methods in Engineering, Mechanical Design, Linear Controls, Experimental Techniques

EXPERIENCE

Gouvis Engineering Consulting Group, Inc.

Irvine, CA

Plumbing Engineering Intern

June 2026 – Sept. 2026

- Designing plumbing systems for residential and commercial buildings by laying out water distribution, drainage/waste/vent, and gas piping in AutoCAD and Revit, producing coordinated construction-ready plan sets
- Performing pipe sizing and fixture-unit calculations against the California Plumbing Code and Title 24 to ensure compliant, efficient system designs
- Coordinating with mechanical, electrical, and architectural teams to resolve conflicts and integrate plumbing within multi-disciplinary projects

HVAC Engineering Intern

June 2025 – Sept. 2025

- Supported HVAC system design by performing load calculations and applying ASHRAE standards and engineering codes, producing and reviewing mechanical plan sets for residential and commercial buildings
- Collaborated with engineers and architects across multi-disciplinary projects, using AutoCAD and Bluebeam Revu for drafting and system layouts

Vertical Flight Society

Nov. 2024 – June 2025

Embedded & Electronics Lead

San Diego, CA

- Led electrical system integration for an eVTOL firefighting drone by selecting batteries, configuring ESCs, setting up a Pixhawk 4 flight controller, and designing the wiring harness, delivering a reliable power architecture for the platform
- Designed, assembled, and tested payload delivery mechanisms by prototyping and iterating on structural components, achieving the reliability needed for real-world firefighting applications
- Coordinated with mechanical and controls teams to optimize power distribution, weight, and system safety across the integrated vehicle
- Mentored new members through embedded systems fundamentals and hardware/software debugging

PROJECTS

Robot Competition: Design & Build

Los Angeles, CA

Mechanical Design Project

Sept. 2024 – Nov. 2024

- Designed, built, and tested a fully functional robot from scratch, engineering the drivetrain and arm/gripper mechanisms to perform reliably under competition conditions
- Modeled 10+ mechanical components in Fusion 360 and produced 2D layouts in AutoCAD, calculating gear ratios and power transmission to improve drivetrain efficiency under load
- Fabricated parts using a laser cutter, drills, taps, and reamers, then assembled the full structure and mechanisms and iterated on the design through testing to refine performance

Thermal & Fluid Dynamics Analysis

San Diego, CA

Experimental & Computational Study

Jan. 2026 – Mar. 2026

- Characterized convective heat-transfer performance by designing and running experiments on an aluminum plate and pin-fin heat sink across free, laminar, and turbulent regimes, fitting a power-law correlation to the Nusselt–Reynolds relationship
- Built Fusion 360 thermal simulations to validate pin-fin geometry, identifying a V-shaped layout that raised the Nusselt number up to ~80% over a bare plate and measurably lowered surface temperature
- Ran ANSYS Fluent CFD on an airfoil and built a MATLAB image-processing pipeline to extract streamlines, validating potential-flow theory to within 1% and quantifying a leading-edge slat that delayed stall from ~12° to ~20°

SKILLS

CAD: SolidWorks, Fusion 360, Creo Parametric, CATIA, AutoCAD, ANSYS, CFD, GD&T

Fabrication: 3D Printing, Laser Cutting, Machining, Sheet Metal Design, Plastic Injection Molding, Soldering, Prototyping

Electromechanical & Programming: Circuit Design, Arduino, Flight Controllers (Pixhawk), Power Distribution, MATLAB, Python